Princewill Okoroafor

Updated October 23, 2026

Email: pco9@cornell.edu **Website**: pokoroafor.github.io

Research interests: Machine Learning Theory, Theoretical CS, Algorithmic Fairness

Current Employer

Harvard SEAS, Responsible ML Foundations — Cambridge, MA
Postdoctoral Researcher with Cynthia Dwork — Sep 2025 - Present

Education

Cornell University — Ithaca, NY
PhD in Computer Science — August 2020 – August 2025
Advisor: Robert Kleinberg
Thesis: *Algorithmic Foundations of Trustworthy Machine Learning*

Harvey Mudd College — Claremont, CA
BS in Computer Science and Mathematics — August 2016 – May 2020

Honors and Scholarships

Cornell Computer Science PhD Dissertation Award	2026
2024 Stanford MS&E Rising Stars	2024
2024 Georgia Tech ISyE Young Researchers	2024
LinkedIn-Cornell 2023 Fellowship Recipient	2023
JP Morgan Fellowship Award Recipient (declined)	2023
Computer Science Department Honors (Harvey Mudd)	2020
ACM ICPC Southern California Regional (5th Place)	2019
Shamit Grover ’05 and Kara McManus Scholarship (Harvey Mudd)	2016
Pan-African Science Olympiad (Gold Medal, First Place)	2015
International Math Olympiad (Honorable Mention)	2012

Industry and Research Experience

Google, Learning Theory — Berlin
Student Researcher with Mehryar Mohri and Julian Zimmert — Winter 2023

Apple, Machine Learning Research — Cupertino, CA
Research Intern with Parikshit Gopalan — Summer 2023

TTIC, Toyota Technological Institute — Chicago, CA
Research Intern with Avrim Blum — Summer 2022

Microsoft Research, MSR NYC — Virtual
Research Engineer Intern with Akshay Krishnamurthy — Summer 2020

Microsoft, Microsoft Teams — Bellevue, WA
Software Engineer Intern — Summer 2018, Summer 2019

Jobs in Nigeria, Lagos, Nigeria
Software Developer Intern Summer 2014

Teaching experience

Teaching assistant, Cornell University 2021 - 2024
CS 6828: Foundations of Responsible Machine Learning
CS 6789: Foundations of Reinforcement Learning
CS 4820: Introduction to Analysis of Algorithms

Teaching assistant, Harvey Mudd College Fall 2017 - Spring 2020
MATH 055: Discrete Mathematics
MATH 171: Abstract Algebra
MATH 172: Abstract Algebra II: Representation Theory
CS 81: Computability and Logic
CS 140: Algorithms
CS 181D: Complexity Theory
PHYS 24: Mechanics and Wave Motion

Publications

Sample-Optimal Multi-Group Learning with an ERM Oracle
Princewill Okoroafor
Preprint, 2026.

The Sample Complexity of Multicalibration with an ERM Oracle
Princewill Okoroafor
Preprint, 2026.

Fast Rates for Swap-Agnostic Learning of Proper Losses
Princewill Okoroafor
Preprint, 2026.

Oracle-Efficient Hybrid Learning with Constrained Adversaries
Princewill Okoroafor, Robert Kleinberg, Michael Kim.
ICLR 2026.

Efficient Opportunistic Approachability
Teodor Marinov, Mehryar Mohri, Princewill Okoroafor, Jon Schneider, Julian Zimmert.
ALT 2026.

Contextual Dynamic Pricing with Heterogeneous Buyers
Thodoris Lykouris, Sloan Nietert, Princewill Okoroafor, Chara Podimata, Julian Zimmert.
NeurIPS 2025.

Near-Optimal Algorithms for Omniprediction

Princewill Okoroafor, Robert Kleinberg, Michael Kim.
FOCS 2025.

Breaking the $T^{2/3}$ Barrier for Sequential Calibration
Yuval Dagan, Constantinos Daskalakis, Maxwell Fishelson, Noah Golowich, Robert Kleinberg, Princewill Okoroafor.
STOC 2025.

Full Swap Regret and Discretized Calibration
Maxwell Fishelson, Robert Kleinberg, Princewill Okoroafor, Renato Paes Leme, Jon Schneider, Yifeng Teng.
ALT 2025.

Omnipredictors for Regression and the Approximate Rank of Convex Functions
Parikshit Gopalan, Princewill Okoroafor, Prasad Raghavendra, Abhishek Shetty, Mihir Singhal.
COLT 2024.

On the Vulnerability of Fairness Constrained Learning to Malicious Noise
Avrim Blum, Princewill Okoroafor, Aadirupa Saha, Kevin Stangl.
AISTATS 2024.

Faster Recalibration of an Online Predictor via Approachability
Princewill Okoroafor, Wen Sun, Robert Kleinberg.
AISTATS 2024.

Non-Stochastic CDF Estimation Using Threshold Queries
Princewill Okoroafor, Vaishnavi Gupta, Robert Kleinberg, Eleanor Goh.
SODA 2023.

Selected Talks

Calibration Measures for Downstream Decision Making Dec 2025
Invited talk, WINE 2025 tutorial on Information Design Perspective on Calibration

Calibration and Trustworthy Decision Making Jun 2025
COLT 2025 Workshop on Predictions and Uncertainty

Near-Optimal Algorithms for Omniprediction Jun 2025
FORC 2025, Stanford University

Near-Optimal Algorithms for Omniprediction Apr 2025
Cornell Theory Seminar

Breaking the $T^{2/3}$ Barrier for Sequential Calibration Oct 2024
FOCS Workshop on Calibration 2024

Omniprediction for Regression and the Approximate Rank of Convex Functions Jul 2024
COLT 2024

Omniprediction for Regression and the Approximate Rank of Convex Functions Dec 2023
Google Learning Theory Seminar

Non-stochastic CDF Estimation Using Threshold Queries Oct 2023
2023 INFORMS Annual Meeting

Non-stochastic CDF Estimation Using Threshold Queries Nov 2022
Cornell Fall 2022 Theory Seminar

Skills

Programming
Proficient in: Python, Javascript, C++, HTML/CSS, Git
Familiar with: Racket, React, AngularJS, NodeJS

Languages
English (fluent), Igbo (fluent), French (conversational)

Service and outreach

Young Data Scientists League Jan 2018 – present
Data science education non-profit focused on middle-school students

Special Maths Academy Nigeria Jan 2020 – August 2024
Math summer program for middle and high school students in Nigeria

Gear Up - Renton, WA Aug 2019
College and career readiness of low-income middle school students.

MISE CS Research Camp - Accra, Ghana Summer 2017
Research mentorship camp for high school students

Gifted Math Camp - Abuja, Nigeria Summer 2015, Summer 2017
High school math camp for exceptional students from across Nigeria.

Other interests

soccer, chess.

References

Robert Kleinberg robert.kleinberg@gmail.com
Professor at Cornell University

Cynthia Dwork dwork@seas.harvard.edu
Gordon McKay Professor of Computer Science at Harvard University

Michael P. Kim mpk@cs.cornell.edu
Assistant Professor at Cornell University

Parikshit Gopalan parik.g@gmail.com
Researcher at Apple